

FocusFlow

AI-Powered Productivity & Study Planner

Full-Stack Web App	Claude API Powered	SaaS / Freemium	Students + Corporate
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Complete Founder's Blueprint | April 2026

Prepared by the FocusFlow founding team. This document is confidential.

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1. Executive Summary

FocusFlow is an AI-powered, full-stack web application designed to become the most intelligent productivity and study planner on the market. It serves two primary audiences — students managing academic workloads and corporate professionals juggling deadlines and projects — and supercharges their daily planning with the Anthropic Claude API.

At its core, FocusFlow eliminates the two biggest reasons people fail at personal productivity: **goal paralysis** and **motivation decay**. The AI Goal Breakdown engine (powered by Claude Sonnet 4.6) converts any large ambition into a clear, daily action plan. The Pomodoro Timer with streaks keeps focus alive. The Weekly AI Digest holds users accountable and continuously improves their strategy.

Claude was chosen over other AI providers for its superior 200K token context window (ideal for the Document Summariser), exceptional instruction-following for structured JSON output, and its Constitutional AI safety layer — important for handling student and professional content responsibly.

Product Name	FocusFlow — AI Productivity & Study Planner
Type	Full-Stack SaaS Web Application (PWA)
AI Engine	Anthropic Claude API — Sonnet 4.6 + Haiku 4.5
Monetisation	Freemium + Pro Subscription (£7.99/mo) + Institution Licensing
Total Dev Time	16 Weeks (4 months) for full MVP launch
Monthly Dev Cost	~£15/month (Claude API dev usage only; hosting is free tier)
Target Launch	August 2026

2. Problem Statement

Students

- Overwhelmed by multiple subjects, assignment deadlines, and exam schedules simultaneously.
- No personalised study plan — generic to-do apps don't understand academic context.
- Long documents (PDFs, lecture notes) are time-consuming to review.
- No insight into how productively they are using their study time.

Corporate Workers

- Big quarterly goals rarely translate into concrete daily actions.
- No AI-assisted weekly retrospective to course-correct strategy.
- Meeting overload reduces deep work time with no tracking.
- Scattered across 5+ tools with no unified productivity hub.

Existing tools like Notion, Todoist, and Google Calendar are powerful but **generic**. None offer AI-native planning, document intelligence, or adaptive weekly coaching. **FocusFlow fills this gap — powered by Claude.**

3. Solution Overview — What is FocusFlow?

FocusFlow is a single, beautifully designed web application that combines seven powerful modules into one seamless experience. Every feature is connected — tasks feed analytics, focus sessions update streaks, uploaded documents generate flashcards, and Claude reads all of it at week's end to deliver personalised coaching.

- **Auth System** — Secure JWT-based registration, login, and session management.
- **Task Manager** — Full CRUD tasks with priorities, categories, due dates, and status tracking.
- **AI Goal Breakdown** — Type any goal — Claude Sonnet 4.6 returns a structured daily plan auto-saved as tasks.
- **Document Summariser** — Upload PDFs; Claude returns a summary, key bullets, and Q&A; flashcards.
- **Pomodoro Focus Timer** — 25/5 cycles with session logging, productivity streaks, and desktop notifications.
- **Analytics Dashboard** — Charts showing task completion rates, focus hours, and weekly performance.
- **Weekly AI Digest** — Every Sunday, Claude reviews your week and delivers personalised coaching.

4. Target Users & Market Opportunity

Global Student Population	~220 million university students worldwide
Remote/Hybrid Workers	~1.3 billion knowledge workers globally
Productivity App Market (2024)	USD \$102 billion — growing at 13% CAGR
AI Productivity Tools (2026)	Projected USD \$24 billion sub-segment
Addressable Niche (Year 1)	50,000 users (students + junior professionals)
Realistic Paid Conversion	5–10% → 2,500–5,000 paying users in Year 1

5. Core Features — Detailed Specification

Feature 1: User Authentication

- JWT access token (15 min) + refresh token (7 days), bcrypt password hashing.
- httpOnly cookie storage — XSS protected.
- Optional Google OAuth 2.0 sign-in.
- Password reset via SendGrid email.

Feature 2: Task Manager

- Create tasks: title, description, due date, priority (High/Medium/Low), category, status.
- Drag-and-drop Kanban board (To-Do / In-Progress / Done).
- Bulk actions, sub-tasks, recurring tasks.

Feature 3: AI Goal Breakdown — Claude Sonnet 4.6

- User enters a high-level goal with optional deadline and context.
- Claude Sonnet 4.6 uses chain-of-thought reasoning to return: phases, weekly milestones, daily actions.
- Generated plan auto-saved as tasks. User can regenerate, edit, or accept individual steps.
- Rate-limited per tier (3/month free · unlimited Pro).

Feature 4: Document Summariser — Claude Sonnet 4.6 (200K Context)

- Upload PDF or paste text — Claude's 200K context window handles full textbooks without chunking.
- Returns: 300-word summary, 5–10 key bullet points, and Q&A; flashcards.
- Flashcard study mode with flip-card UI. Export to PDF or Anki format.

Feature 5: Pomodoro Focus Timer

- Classic 25/5 cycles with customisable intervals (Pro).
- Each session logged with linked task, duration, and timestamp.
- Productivity streak counter + focus heatmap.

Feature 6: Analytics Dashboard

- Weekly completion bar chart, category pie chart, GitHub-style focus heatmap.
- Month-over-month productivity trend line chart.

Feature 7: Weekly AI Digest — Claude Sonnet 4.6

- node-cron job runs every Sunday at 8:00 PM.
- Claude reads: tasks completed, focus hours, missed tasks, category breakdown.
- Returns personalised coaching: what went well, what to improve, 3 specific next-week recommendations.
- Delivered in-app + optional SendGrid email. Digest history saved for tracking.

6. Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + TypeScript	Component-based UI with strict typing
Frontend	Tailwind CSS + shadcn/ui	Rapid, consistent, accessible styling
Frontend	Zustand + React Query	Global state + server state caching
Frontend	Recharts	Analytics charts and visualisation
Backend	Node.js + Express.js	Stateless REST API with versioned routes
Backend	JWT + bcrypt	Authentication and password security
Backend	Multer + node-cron	File uploads + scheduled cron jobs
Backend	SendGrid	Transactional emails + Weekly Digest delivery
Database	PostgreSQL + Prisma ORM	Relational DB with type-safe queries
Database	Supabase (managed)	Free-tier hosted PostgreSQL
AI Layer	Anthropic Claude API	claude-sonnet-4-6 (complex AI) + claude-haiku-4-5 (fast/cheap tasks)
AI Layer	LangChain.js	Document loading, chunking, Claude chains
Infra	Docker + Render	Containerised dev + free-tier production
Infra	Cloudinary / AWS S3	PDF and file storage with signed URLs
Infra	GitHub Actions	CI/CD: test → build → deploy
Testing	Jest + Supertest	Backend unit and integration tests
Testing	Vitest + RTL	Frontend component tests
Billing	Stripe	Pro subscriptions + team plan billing

Claude API Model Routing Strategy

Feature	Model	Reason
AI Goal Breakdown	claude-sonnet-4-6	Deep reasoning + strict structured JSON output
Document Summariser	claude-sonnet-4-6	200K token context — handles full textbooks natively
Flashcard Generation	claude-haiku-4-5	Fast & cheap — simple Q&A; format, high volume

Weekly AI Digest	claude-sonnet-4-6	Personalised coaching quality matters here
Task suggestions / tips	claude-haiku-4-5	Low complexity — keeps API costs minimal

7. System Architecture Overview

FocusFlow follows a clean three-tier architecture: **Client** → **API Server** → **Database + Claude API + Storage**.

Client (React PWA)

React SPA served via CDN. Communicates with backend via REST API using Axios. JWT stored in httpOnly cookies.

API Server (Express.js on Render)

Stateless REST API with versioned routes (/api/v1/...). Handles auth, task CRUD, file uploads, Claude proxy calls, and cron scheduling.

Database (PostgreSQL on Supabase)

Managed PostgreSQL. Prisma ORM for all queries. Tables: users, tasks, pomodoro_sessions, documents, flashcards, digests.

Claude AI Service Layer

Node.js service module wrapping the Anthropic SDK. Routes requests to claude-sonnet-4-6 or claude-haiku-4-5 based on task complexity. Rate-limited and response-cached.

File Storage (Cloudinary / S3)

Uploaded PDFs stored with signed URLs for secure retrieval by the Claude Document Summariser.

Cron Jobs

node-cron runs inside the server. Weekly Digest job queries DB, calls Claude Sonnet 4.6, fires SendGrid email every Sunday at 8 PM.

8. Step-by-Step Execution Plan (16 Weeks)

Phase	Duration	Key Deliverables
Phase 1 Foundation (Weeks 1–2)	Weeks 1–2	• Monorepo structure (client/ + server/) • TypeScript, ESLint, Prettier config • PostgreSQL + Prisma schema • Docker Compose + GitHub repo + CI
Phase 2 Auth (Weeks 3–4)	Weeks 3–4	• Register/Login/Logout API • JWT + refresh token logic • Password reset via SendGrid • React auth screens + protected routes
Phase 3 Task Manager (Weeks 5–6)	Weeks 5–6	• Full CRUD task API • Kanban drag-and-drop board • Sub-tasks + recurring tasks • Integration tests
Phase 4 Claude AI Goals (Weeks 7–8)	Weeks 7–8	• Anthropic SDK integration • claude-sonnet-4-6 prompt engineering • Goal input → task auto-creation • Per-tier rate limiting
Phase 5 Doc Summariser (Weeks 9–10)	Weeks 9–10	• Multer file upload + Cloudinary • LangChain + Claude 200K context • Summary + flashcard generation • Study mode UI + Anki export
Phase 6 Pomodoro + Analytics (Weeks 11–12)	Weeks 11–12	• Pomodoro timer + session logging • Streak calculation logic • Analytics aggregation endpoints • Recharts dashboard UI
Phase 7 Weekly Digest (Weeks 13–14)	Weeks 13–14	• node-cron Sunday job • Claude Sonnet 4.6 coaching prompt • SendGrid email template • In-app notification + digest history
Phase 8 Launch (Weeks 15–16)	Weeks 15–16	• Full test coverage (Jest + Vitest) • Performance + code splitting • Render production deploy • Stripe billing + Product Hunt launch

9. Everything You Need to Build FocusFlow

Skills / Knowledge

- JavaScript / TypeScript (intermediate–advanced)
- React fundamentals + Node.js + Express REST API design
- SQL basics and Prisma ORM usage
- Basic understanding of JWT and auth flows
- Anthropic Claude API / prompt engineering (learn as you go)

Accounts & API Keys

- Anthropic Console account — Claude API key (pay-per-use, ~£15/month for dev)
- SendGrid account — free tier (100 emails/day)
- Cloudinary account — free tier for file storage
- Render account — free tier for backend + frontend hosting
- Supabase account — free tier for managed PostgreSQL
- Stripe account — for Pro subscription billing
- GitHub account — repo hosting + Actions CI/CD

Software Tools (You Already Have Most of These)

- VS Code ✓ | Docker Desktop ✓ | Postman ✓ | Windows 11 ✓
- Node.js v20+, Git, PostgreSQL client (DBeaver or TablePlus)

Monthly Costs (Dev Phase — 0 to 100 users)

- Render free tier: £0 | Supabase free tier: £0 | SendGrid free tier: £0
- Cloudinary free tier: £0 | Anthropic Claude API (dev): ~£15/month
- TOTAL: ~£15/month to build and run the full MVP

10. How People Will Access FocusFlow

FocusFlow is a **Progressive Web Application (PWA)** — accessed via any modern web browser and installable on any device like a native app.

Browser (Desktop & Mobile)

Visit focusflow.app in Chrome, Firefox, Safari, or Edge. Full functionality, zero install.

Installed PWA

Add to home screen on iOS/Android or desktop Chrome. Launches standalone with offline support.

Email Notifications

Weekly AI Digest every Sunday. Password reset. Optional daily task reminders via SendGrid.

Push Notifications

Pomodoro alerts, deadline reminders, streak notifications via Web Push API.

Future: React Native App

Post-MVP — same Express API reused for iOS + Android native app (Month 11 of roadmap).

11. Web App vs. Native App — The Decision

Criterion	Web App (PWA) ■	Native Mobile App
Build time	Faster — 1 codebase	Slower — separate iOS + Android
Distribution	No app store needed	App Store + Play Store approval
Launch cost	Near zero	\$99/year Apple Developer + time
Cross-platform	All devices	Platform-specific builds
Offline support	Yes — Service Workers	Yes — native
Push notifications	Yes — Web Push API	Yes — native
SEO / Discovery	Yes — Google-indexable	No
Recommended MVP	■ YES	Post-MVP milestone

Verdict: Build FocusFlow as a PWA first. Faster, cheaper, accessible on every device. Add a React Native mobile app as a post-launch milestone at Month 11.

12. Revenue Model & Pricing

Pricing Tiers

Free	£0/month	5 active tasks · 1 AI goal/month · 3 Pomodoros/day · Basic analytics
Pro	£7.99/month	Unlimited tasks & AI · Document summariser · Weekly Digest · Full analytics
Team	£19.99/user/mo	All Pro + team workspaces · manager dashboard · bulk invite · priority support

Revenue Streams

Revenue Stream	Target Segment	Est. Monthly	Notes
Pro Subscriptions	Students & freelancers	£7.99/user/mo	2,500 users × £7.99 = £19,975/mo at scale
Team Licenses	SMEs & remote teams	£19.99/user/mo	50 teams × 5 users = £4,997/mo at scale
University Licensing	HEIs & schools	£500–£2,000/yr	Bulk institutional access deal
Corporate Partnerships	HR / L&D; departments	Custom pricing	White-label deal for onboarding tools

Founder Earnings Projection

Month 1–3 (Beta)	£0 — building audience, collecting feedback
Month 4–6 (100 Pro)	~£799/month — covers Claude API + hosting costs
Month 7–9 (500 Pro)	~£3,995/month — salary-level income
Month 10–12 (1,500 Pro)	~£11,985/month — strong SaaS revenue
Year 2 (5,000 Pro)	~£23,000+/month — sustainable business

13. Benefits to Students

Instant AI Study Plans

Type 'Pass my chemistry final in 3 weeks' — Claude Sonnet 4.6 returns a day-by-day revision plan in seconds.

PDF Summariser Saves Hours

A 200-page textbook becomes a 500-word summary + 20 flashcards in under 60 seconds thanks to Claude's 200K context.

Pomodoro Keeps Focus Honest

Evidence-backed study cycles tracked over time — students see exactly how many hours they studied per subject.

Streak Gamification

Daily study streaks are surprisingly motivating. Same psychology as Duolingo but applied to real academic work.

Free Weekly AI Coaching

The Sunday Digest is like a personal tutor reviewing your week — available to every student, completely free.

Exam Stress Reduction

Organised plans + small daily steps dramatically reduce the psychological burden of exam season.

14. Benefits to Corporate Workers

OKRs → Daily Actions

Claude converts vague quarterly goals like 'Grow Q2 revenue 15%' into concrete weekly and daily tasks.

Deep Work Tracking

Pomodoro session logging gives professionals hard data on focus hours — valuable for WFH accountability.

Document Intelligence

Upload a 40-page board report — get an executive summary in seconds. Claude handles length effortlessly.

Private Weekly Retrospective

Sunday Digest catches strategic drift before it becomes a crisis — automatically, without a manager.

Task Priority Clarity

High/Medium/Low tags always answer 'what is the single most important thing to do right now?'

One Hub — Five Tools Replaced

FocusFlow replaces separate Pomodoro apps, note tools, task managers, and AI chat tools in one login.

15. Benefits to Society

- Democratizes AI coaching: Students from low-income backgrounds get Claude-powered academic guidance for free.
- Reduces cognitive overload: Structured planning reduces anxiety, burnout, and decision fatigue — real mental health benefits.
- Supports neurodivergent users: Structured Pomodoro system, visual boards, and Claude's step-by-step plans benefit users with ADHD or dyslexia.
- Accelerates lifelong learning: The Document Summariser + Flashcard generator makes self-education dramatically faster.
- Reduces educational inequality: A student anywhere in the world gets the same quality of AI-powered planning as one at a top institution.

16. Competitive Landscape

Competitor	Strengths	Gap vs FocusFlow
Notion	Flexible, powerful, popular	No AI goals, no Pomodoro, no digest
Todoist	Clean task management	No AI, no doc summariser, no focus timer
Motion	AI scheduling	Expensive, no study features, no document AI
MyStudyLife	Student-specific calendar	No AI, no focus timer, no document tools
Forest / Be Focused	Pomodoro gamification	Single-feature, no AI, no task management
FocusFlow + Claude ■	All of the above — unified + AI	New entrant — needs brand building

17. Risk Analysis & Mitigation

Risk	Likelihood	Impact	Mitigation
Claude API cost spiral	High	Medium	Cache responses, rate-limit per user tier, route simple tasks to claude-haiku-4-5 (much cheaper).
Low free→paid conversion	Medium	Medium	14-day Pro trial, in-app nudges at natural friction points, compelling onboarding.
Big tech builds same feature	Low	High	Niche focus (students + corporate) and community loyalty are the moat. Move fast.
Solo dev burnout	Medium	High	Follow the 16-week phased plan strictly. One feature per phase. Avoid scope creep.
Data privacy concerns	Medium	Low	GDPR compliance, clear privacy policy, encrypted storage, no selling user data.
Low organic discovery	Medium	Medium	Student communities (Reddit, Discord, TikTok) + SEO-optimised landing page.

18. Launch Strategy & Marketing

Pre-Launch (Weeks 13–16)

- Landing page with email waitlist (Mailchimp or ConvertKit).
- Build-in-public weekly updates on Twitter/X and LinkedIn with screenshots.
- Student subreddits (r/GetStudying, r/productivity) — contribute genuinely.
- 2-minute demo video on TikTok, YouTube Shorts, Instagram Reels.
- Reach out to university student unions for beta testers.

Launch Week

- Product Hunt (Tuesday, 12:01 AM PST).
- Personal founder email to entire waitlist.
- Indie Hackers story: 'I built a Claude-powered study app in 16 weeks'.
- First 100 sign-ups: lifetime 50% off Pro.

Post-Launch Growth

- SEO content: 'Best AI study planner 2026', 'Claude API productivity tools'.
- Partner with student YouTubers and productivity creators for reviews.
- Referral programme: 1 free Pro month per referred paid user.
- University bulk licensing pitch.

19. 12-Month Growth Roadmap

Months 1–4	MVP Build	All 7 features built, internal testing, private beta with 20 users
Month 5	Public Beta	Product Hunt, waitlist email, free tier open to all
Month 6	Pro Launch	Stripe billing live — target 100 paying users
Months 7–8	Feature Polish	Performance, mobile PWA, user-requested features
Month 9	Team Plan	Workspace sharing, manager dashboard, bulk invite
Month 10	University Outreach	First institutional licensing deal closed
Month 11	React Native App	iOS + Android app on app stores
Month 12	1,000 Pro Users	Sustainable MRR — consider hiring first part-time developer

20. Conclusion & Call to Action

FocusFlow is not just another to-do list. It is a category-defining AI productivity companion powered by the Anthropic Claude API — the most capable and context-rich LLM available. Every design decision — freemium pricing, PWA architecture, Claude Sonnet 4.6 for heavy AI tasks, claude-haiku-4-5 for lightweight tasks — has been made to maximise user value and sustainable revenue for you as the founder.

The competitive window is open. AI-native productivity tools are a nascent category. FocusFlow with Claude API can be live, subscription-funded, and growing within 5 months.

Start Week 1 today. Build the repo. Ship the auth. The rest follows.